TakeMeter — Fine-Tuning Starter Notebook

AI201 · Project 3

This notebook walks you through fine-tuning a text classifier on your annotated dataset and comparing it to a zero-shot baseline.

What this notebook does for you (infrastructure):

- Tokenizes your dataset and prepares it for training
- Runs the fine-tuning pipeline with DistilBERT
- Computes evaluation metrics and generates a confusion matrix
- Runs the Groq baseline and compares both models

What you do (the actual work):

- Collect and annotate your 200+ examples (done before opening this notebook)
- Define your label map and upload your CSV
- Write your Groq classification prompt using your label definitions
- Analyze the output and write your evaluation report

Before you start: Make sure you are using a T4 GPU runtime.
Go to **Runtime → Change runtime type → T4 GPU**, then click Save.

```
# Install any dependencies not pre-installed on Colab
!pip install -q groq python-dotenv
print("✅ Dependencies ready")
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143.7/143.7 kB 8.8 MB/s eta 0:00:00
✅ Dependencies ready
```

```
import pandas as pd
import numpy as np
import json
import time

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, classification_report,
    confusion_matrix, ConfusionMatrixDisplay,
)
import matplotlib.pyplot as plt

import torch
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
    TrainingArguments,
    Trainer,
    DataCollatorWithPadding,
)
from datasets import Dataset
import warnings
warnings.filterwarnings("ignore")

print("✅ Imports complete")
print(f"PyTorch version: {torch.__version__}")
print(f"GPU available: {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
```

```
✅ Imports complete
PyTorch version: 2.11.0+cu128
GPU available: True
GPU: Tesla T4
```

Section 1: Load Your Dataset

Upload your labeled CSV and define your label map.

Your CSV must have at least two columns: `text` (the post/comment) and `label` (your string label).

```
# —— TODO ————————————————————————————————————————————————————————————————
# Define YOUR label map below.
# Keys are the string labels in your CSV; values are integers starting at 0.
# Add or remove entries to match your actual labels (2–4 labels supported).
#
# The example below is ILLUSTRATIVE ONLY (the r/nba taxonomy from the project
# page). DELETE it and use your own community's labels — submitting the
# example unchanged will not pass.
# ————————————————————————————————————————————————————————————————————————

LABEL_MAP = {
    "analysis":  0,   # ← Replace with your first label
    "hot_take":  1,   # ← Replace with your second label
    "reaction":  2,   # ← Replace with your third label (remove if you have 2 labels)
    # "fourth_label": 3,  # ← Uncomment if you have a fourth label
}

# —— END TODO ————————————————————————————————————————————————————————————

ID_TO_LABEL = {v: k for k, v in LABEL_MAP.items()}
NUM_LABELS = len(LABEL_MAP)
print(f"Labels: {LABEL_MAP}")
print(f"Number of labels: {NUM_LABELS}")
```

```
Labels: {'analysis': 0, 'hot_take': 1, 'reaction': 2}
Number of labels: 3
```

```
# Upload your CSV from your computer
from google.colab import files
print("Select your labeled dataset CSV file...")
uploaded = files.upload()
CSV_PATH = list(uploaded.keys())[0]
print(f"Uploaded: {CSV_PATH}")
```

```
Select your labeled dataset CSV file...
```

Choose Files takemeter.csv

takemeter.csv(text/csv) - 138126 bytes, last modified: 6/23/2026 - 100% done

```
Saving takemeter.csv to takemeter.csv
Uploaded: takemeter.csv
```

```
# Load and validate your dataset
df = pd.read_csv(CSV_PATH)

# —— TODO (if needed) ————————————————————————————————————————————————————
# If your CSV uses different column names, rename them here.
# Example: df = df.rename(columns={"post": "text", "category": "label"})
# —— END TODO ————————————————————————————————————————————————————————————

print(f"Columns: {df.columns.tolist()}")
print(f"Total examples: {len(df)}")
print()
print("Label distribution:")
print(df["label"].value_counts())

# Validate all labels are in LABEL_MAP
unknown = set(df["label"].unique()) - set(LABEL_MAP.keys())
if unknown:
    print(f"\n⚠️  Labels in CSV not found in LABEL_MAP: {unknown}")
    print("Update your LABEL_MAP above to include all labels.")
else:
    print("\n✅ All labels match your LABEL_MAP")

# Convert string labels to integers
df["label_id"] = df["label"].map(LABEL_MAP)
df = df.dropna(subset=["label_id"])
df["label_id"] = df["label_id"].astype(int)
```

```
Columns: ['text', 'label', 'notes', 'source', 'permalink', 'prelabeled']
Total examples: 219

Label distribution:
label
```

```
analysis    80
reaction    70
hot_take    69
Name: count, dtype: int64

✅ All labels match your LABEL_MAP
```

Section 2: Prepare Data for Training

Splits your dataset into train / validation / test sets and tokenizes the text.

```
# Train / val / test split — 70% / 15% / 15%
# Stratified so each split has roughly the same label distribution.
train_df, temp_df = train_test_split(
    df, test_size=0.30, random_state=42, stratify=df["label_id"]
)
val_df, test_df = train_test_split(
    temp_df, test_size=0.50, random_state=42, stratify=temp_df["label_id"]
)

print(f"Train: {len(train_df)} examples")
print(f"Validation: {len(val_df)} examples")
print(f"Test: {len(test_df)} examples")
print()
print("Train label distribution:")
print(train_df["label"].value_counts())
print()
print("Test label distribution:")
print(test_df["label"].value_counts())

# Reset indices (needed for clean HuggingFace Dataset conversion)
train_df = train_df.reset_index(drop=True)
val_df   = val_df.reset_index(drop=True)
test_df  = test_df.reset_index(drop=True)
```

```
Train: 153 examples
Validation: 33 examples
Test: 33 examples

Train label distribution:
label
analysis    56
reaction    49
hot_take    48
Name: count, dtype: int64

Test label distribution:
label
analysis    12
hot_take    11
reaction    10
Name: count, dtype: int64
```

```
# Load tokenizer and tokenize all splits
MODEL_NAME = "distilbert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)

def tokenize(examples):
    return tokenizer(examples["text"], truncation=True, max_length=256)

def make_dataset(df_split):
    ds = Dataset.from_pandas(
        df_split[["text", "label_id"]].rename(columns={"label_id": "labels"})
    )
    return ds.map(tokenize, batched=True)

train_dataset = make_dataset(train_df)
val_dataset   = make_dataset(val_df)
test_dataset  = make_dataset(test_df)

data_collator = DataCollatorWithPadding(tokenizer=tokenizer)
print("✅ Tokenization complete")
print(f"Sample keys: {list(train_dataset[0].keys())}")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_
config.json: 100%  483/483 [00:00<00:00, 38.1kB/s]
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 4.39kB/s]
vocab.txt: 100%  232k/232k [00:00<00:00, 757kB/s]
tokenizer.json: 100%  466k/466k [00:00<00:00, 1.93MB/s]
Map: 100%  153/153 [00:00<00:00, 1545.43 examples/s]
Map: 100%  33/33 [00:00<00:00, 807.67 examples/s]
Map: 100%  33/33 [00:00<00:00, 836.88 examples/s]
✅ Tokenization complete
Sample keys: ['text', 'labels', 'input_ids', 'token_type_ids', 'attention_mask']
```

Section 3: Fine-Tune Your Model

Loads `distilbert-base-uncased` with a classification head and fine-tunes it on your training data.
Training runs for 3 epochs and takes **5–15 minutes** on a T4 GPU.

> **Hyperparameter note:** The defaults below work well for datasets of 100–500 examples.
> If you change any values, note what you changed and why in your README.

```
# Load DistilBERT with a classification head
model = AutoModelForSequenceClassification.from_pretrained(
    MODEL_NAME,
    num_labels=NUM_LABELS,
    id2label=ID_TO_LABEL,
    label2id=LABEL_MAP,
)
print(f"✅ Model loaded: {MODEL_NAME}")
print(f"Output labels: {NUM_LABELS}")
```

```
model.safetensors: 100%  268M/268M [00:03<00:00, 824MB/s]
Loading weights: 100%  100/100 [00:00<00:00, 3008.29it/s]
[transformers] DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_transform.bias    | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
classifier.bias         | MISSING    |
pre_classifier.weight   | MISSING    |
classifier.weight       | MISSING    |
pre_classifier.bias     | MISSING    |

Notes:
- UNEXPECTED:  can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstr
✅ Model loaded: distilbert-base-uncased
Output labels: 3
```

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return {"accuracy": accuracy_score(labels, predictions)}
```

```
# ── Hyperparameters ──────────────────────────────────────────────
# num_train_epochs  — passes through the training data; 3 is a good default
#                     for small datasets. Increase cautiously; more epochs
#                     risk overfitting on 200 examples.
# learning_rate     — 2e-5 is the standard starting point for fine-tuning
#                     BERT-family models. Lower → slower but more stable.
# per_device_train_batch_size — 16 fits T4 GPU comfortably.
#                     Reduce to 8 if you get out-of-memory errors.
# ─────────────────────────────────────────────────────────────────
```

```
training_args = TrainingArguments(
    output_dir="./takemeter-model",
    num_train_epochs=3,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=32,
    learning_rate=2e-1,
    weight_decay=0.01,
    warmup_steps=50,
    eval_strategy="epoch",
    save_strategy="epoch",
    save_total_limit=1,
    load_best_model_at_end=True,
    metric_for_best_model="accuracy",
    logging_steps=10,
    report_to="none",
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
)

print("Starting fine-tuning... (5-15 minutes on T4 GPU)")
trainer.train()
print("\n✅ Fine-tuning complete")
```

```
Starting fine-tuning... (5-15 minutes on T4 GPU)
```

[30/30 00:28, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy
1	1.684263	1.096729	0.363636
2	1.686892	2.287339	0.363636
3	1.824763	1.160707	0.363636

Writing model shards: 100% 1/1 [00:00<00:00, 1.10it/s]

Writing model shards: 100% 1/1 [00:07<00:00, 7.04s/it]

Writing model shards: 100% 1/1 [00:00<00:00, 1.04it/s]

```
✅ Fine-tuning complete
```

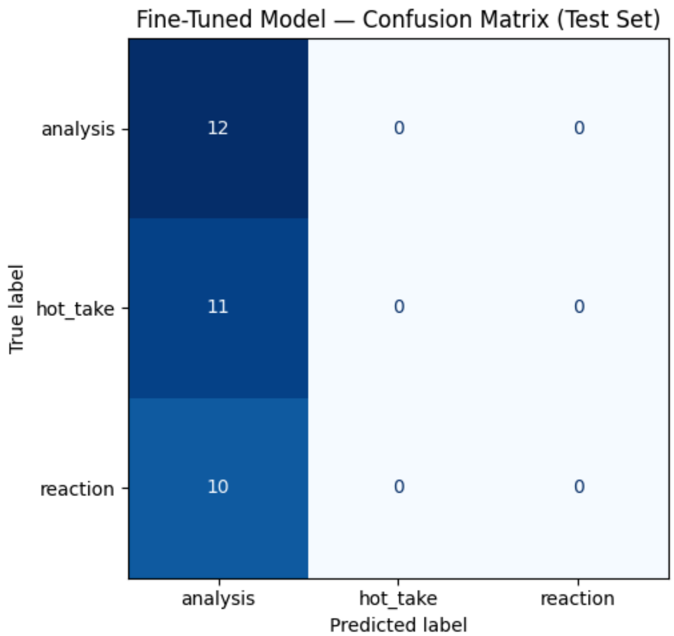
Section 4: Evaluate Fine-Tuned Model on Test Set

Runs inference on your locked test set and generates metrics and a confusion matrix.
These numbers go directly into your evaluation report.

```
# Run inference on the test set
print("Running inference on test set...")
ft_output = trainer.predict(test_dataset)
ft_pred_ids = np.argmax(ft_output.predictions, axis=-1)
ft_true_ids = ft_output.label_ids

ft_probs = torch.nn.functional.softmax(
    torch.tensor(ft_output.predictions), dim=-1
).numpy()

# Overall accuracy
ft_accuracy = accuracy_score(ft_true_ids, ft_pred_ids)
print(f"\n🎯 Fine-tuned model accuracy: {ft_accuracy:.3f}")

# Per-class metrics
label_names = [ID_TO_LABEL[i] for i in range(NUM_LABELS)]
print("\nPer-class metrics (fine-tuned model):")
print(classification_report(ft_true_ids, ft_pred_ids, target_names=label_names, zero_division=0))
```

```
Running inference on test set...

🎯 Fine-tuned model accuracy: 0.364

Per-class metrics (fine-tuned model):
              precision    recall  f1-score   support

    analysis       0.36      1.00      0.53        12
    hot_take       0.00      0.00      0.00        11
    reaction       0.00      0.00      0.00        10

    accuracy                           0.36        33
   macro avg       0.12      0.33      0.18        33
weighted avg       0.13      0.36      0.19        33
```

```
# Confusion matrix
cm = confusion_matrix(ft_true_ids, ft_pred_ids)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=label_names)
fig, ax = plt.subplots(figsize=(7, 5))
disp.plot(ax=ax, cmap="Blues", colorbar=False)
ax.set_title("Fine-Tuned Model — Confusion Matrix (Test Set)")
plt.tight_layout()
plt.savefig("confusion_matrix.png", dpi=150)
plt.show()
print("✅ Saved: confusion_matrix.png  →  commit this to your repo and include in README")
```



✅ Saved: confusion_matrix.png → commit this to your repo and include in README

```
# Print wrong predictions for your error analysis
# Review these carefully — pick 3 to analyze in depth in your README.

wrong_idx = np.where(ft_pred_ids != ft_true_ids)[0]
print(f"Wrong predictions: {len(wrong_idx)} / {len(ft_true_ids)}\n")

for i, idx in enumerate(wrong_idx[:15]):
    text = test_df.iloc[idx]["text"]
    true_label = ID_TO_LABEL[ft_true_ids[idx]]
    pred_label = ID_TO_LABEL[ft_pred_ids[idx]]
    confidence = ft_probs[idx][ft_pred_ids[idx]]
    print(f"--- #{i+1} ---")
    print(f"Text:      {text[:200]}{'...' if len(text) > 200 else ''}")
    print(f"True:      {true_label}")
    print(f"Predicted: {pred_label}  (confidence: {confidence:.2f})")
    print()
```

```
--- #7 ---
Text:      I've always had the mindset that if a person truly claims to hate any one genre entirely then it's not re
True:      hot_take
Predicted: analysis  (confidence: 0.37)

--- #8 ---
Text:      Thta mixed with the vocal "once you free your mind of music being correct..." ahh! I'll never forget the
True:      reaction
Predicted: analysis  (confidence: 0.37)

--- #9 ---
Text:      The thunderbolt I experienced the first time I listened to The Wall was unbelievable.  I was 11, and it w
True:      reaction
Predicted: analysis  (confidence: 0.37)

--- #10 ---
Text:      You are allowed to voice your opinion. There is no obligation to preface every single thing you say with

      It sounds more like...
True:      hot_take
Predicted: analysis  (confidence: 0.37)

--- #11 ---
Text:      Black Sabbath was directly influenced by Gustav Holst's 1918 orchestral suite The Planets, so for all int
True:      hot_take
Predicted: analysis  (confidence: 0.37)

--- #12 ---
Text:      I listen to whole albums and definitely associate memories with when I was listening to that album bc I s
True:      reaction
Predicted: analysis  (confidence: 0.37)

--- #13 ---
Text:      Humans are the worlds worst animals driven by narcissistic feelings, emotions, and the primal need to pro

      We completely ignore logic or reason

      Its how were made  ¯_(ツ)_/¯
True:      hot_take
Predicted: analysis  (confidence: 0.37)

--- #14 ---
Text:      They do not, it's just that unless everyone prefaces their opinion with "this is just my opinion", y'all
True:      hot_take
Predicted: analysis  (confidence: 0.37)

--- #15 ---
Text:      I grew up listening to albums so when I hear those particular ones now those memories are still evoked (e
True:      reaction
Predicted: analysis  (confidence: 0.37)
```

Section 5: Baseline Classifier (Groq)

Runs your zero-shot baseline using `llama-3.3-70b-versatile`.
You need to write the classification prompt using your label definitions.

```
from groq import Groq

# —— TODO: Add your Groq API key ———————————————————————————————————————————
# Recommended: use Colab Secrets so your key is never visible in the notebook.
#   1. Click the 🔑 icon in the left sidebar ("Secrets")
#   2. Add a secret named GROQ_API_KEY with your key as the value
#   3. Enable notebook access for the secret
#
# Then uncomment Option A below (and delete Option B).
#
# Option A — Colab Secrets (recommended):
from google.colab import userdata
GROQ_API_KEY = userdata.get("GROQ_API_KEY")
#
# Option B — paste directly (do not commit to GitHub):
#GROQ_API_KEY = "your_groq_api_key_here"

assert GROQ_API_KEY, (
    "GROQ_API_KEY not set — add it in the Colab Secrets panel (\U0001f511, left "
    "sidebar) and enable notebook access for this notebook, or use Option B above."
```

```
)

client = Groq(api_key=GROQ_API_KEY)
print("✅ Groq client initialized")
```

```
✅ Groq client initialized
```

```
# ── TODO: Write your classification prompt ────────────────────────────────
# Your prompt should:
#   1. Name your community and task
#   2. Define each label in plain language (copy from your planning.md)
#   3. Give one example post per label
#   4. Tell the model to output ONLY the label name — nothing else
#
# The model's response must match one of your label strings exactly,
# or the classify_with_groq() function below will mark it as unparseable.
#
# ──────────────────────────────────────────────────────────────────────────
# REPLACE the placeholders below with your actual prompt. As written, this
# skeleton will NOT classify correctly — you must fill it in.

# SYSTEM_PROMPT = """
# You are classifying <posts> from <YOUR COMMUNITY>.
# Assign each post to exactly one of the following categories.

# <label_1>: <one-sentence definition of label 1>
# Example: "<one example post for label_1>"

# <label_2>: <one-sentence definition of label 2>
# Example: "<one example post for label_2>"

# <label_3>: <one-sentence definition of label 3>
# Example: "<one example post for label_3>"

# Respond with ONLY the label name.
# Do not explain your reasoning.

# Valid labels:
# <label_1>
# <label_2>
# <label_3>
# """


SYSTEM_PROMPT = """
  You are classifying comments and posts from r/LetsTalkMusic, a subreddit for in-depth music discussion. Assign
  each one to exactly one category based on HOW it supports its point.

  analysis: a structured argument about the music backed by specific, verifiable evidence
  (production/harmony/rhythm detail, historical or contextual fact, or named song/artist examples); the
  reasoning would stand even if the opinion framing were removed.
  Example: "Shoegaze isn't just a 'wall of noise' — My Bloody Valentine's Loveless uses the glide-guitar
  technique, detuning mid-chord with the tremolo arm, which is what creates that seasick pitch-bend. It's
  harmonically deliberate, not just distortion."

  hot_take: a bold, confident evaluative claim asserted without genuine support; it may sound credible but does
  not actually reason.
  Example: "Radiohead is the most overrated band of all time; people only like OK Computer to seem smart."

  reaction: an immediate emotional or personal response — a feeling or a memory — that makes no claim about the
  music's quality.
  Example: "First time hearing Carrie & Lowell and I'm literally sobbing, I don't have words."

  Decision rules for hard cases:
  - If a fact is cited but it is vague, cherry-picked, or decorative (there to sound credible while delivering a
  put-down), choose hot_take. Choose analysis only if it genuinely reasons.
  - An unsupported judgment about quality (e.g. "worst album ever") is hot_take. A pure feeling or personal
  experience with no quality claim is reaction.
  - If a musical detail only serves a feeling, choose reaction; if a feeling decorates a real argument, choose
  analysis.
  - When in doubt, prefer the lower-evidence label: reaction < hot_take < analysis.

  Respond with ONLY the label name, exactly as written here, using the underscore in hot_take: analysis,
  hot_take, or reaction. Do not explain.

  Valid labels:
  analysis
```

```
  hot_take
  reaction
  """

print("Prompt length:", len(SYSTEM_PROMPT), "characters")
```

```
Prompt length: 1961 characters
```

```
def classify_with_groq(text):
    """Classify a single post. Returns a label string or None if unparseable."""
    try:
        response = client.chat.completions.create(
            model="llama-3.3-70b-versatile",
            messages=[
                {"role": "system", "content": SYSTEM_PROMPT},
                {"role": "user", "content": f"Classify this post:\n\n{text}"},
            ],
            temperature=0,
            max_tokens=20,
        )
        raw = response.choices[0].message.content.strip().lower()
        # Match the model's output to a label. Check longest labels first so a
        # label that is a substring of another (e.g. "recommendation" vs.
        # "strong_recommendation") can't be matched by mistake.
        for label in sorted(LABEL_MAP, key=len, reverse=True):
            if raw == label or label in raw:
                return label
        return None  # model output didn't match any known label
    except Exception as e:
        print(f"API error: {e}")
        return None


# Run baseline on test set
print(f"Running baseline on {len(test_df)} examples...")
print("(May take a few minutes — 0.1s delay between requests to respect free-tier limits)\n")

baseline_preds = []
for i, (_, row) in enumerate(test_df.iterrows()):
    pred = classify_with_groq(row["text"])
    baseline_preds.append(pred)
    if (i + 1) % 10 == 0:
        print(f"  {i+1}/{len(test_df)} complete...")
    time.sleep(0.1)

none_count = baseline_preds.count(None)
if none_count > 0:
    print(f"\n⚠️  {none_count} responses could not be parsed.")
    print("Review your prompt — the model may not be outputting clean label names.")
```

```
Running baseline on 33 examples...
(May take a few minutes — 0.1s delay between requests to respect free-tier limits)

  10/33 complete...
  20/33 complete...
  30/33 complete...
```

```
# Baseline metrics (exclude unparseable responses)
valid = [(p, t) for p, t in zip(baseline_preds, test_df["label_id"])
         if p is not None]
bl_pred_ids = [LABEL_MAP[p] for p, _ in valid]
bl_true_ids = [t for _, t in valid]

bl_accuracy = accuracy_score(bl_true_ids, bl_pred_ids)
print(f"🎯 Baseline accuracy: {bl_accuracy:.3f}  "
      f"(evaluated on {len(valid)}/{len(test_df)} parseable responses)")
print()
label_names = [ID_TO_LABEL[i] for i in range(NUM_LABELS)]
print("Per-class metrics (baseline):")
print(classification_report(bl_true_ids, bl_pred_ids, target_names=label_names, zero_division=0))
```

```
🎯 Baseline accuracy: 0.879  (evaluated on 33/33 parseable responses)

Per-class metrics (baseline):
              precision    recall  f1-score   support

    analysis       1.00      0.92      0.96        12
```

```
    hot_take       0.89      0.73      0.80        11
    reaction       0.77      1.00      0.87        10

    accuracy                           0.88        33
   macro avg       0.89      0.88      0.88        33
weighted avg       0.89      0.88      0.88        33
```

Section 6: Compare Results and Export

Side-by-side comparison of both models.
Download the output files and commit them to your GitHub repo.

```
print("=" * 50)
print("RESULTS COMPARISON")
print("=" * 50)
print(f"{'Model':<35} {'Accuracy':>8}")
print("-" * 45)
print(f"{'Zero-shot baseline (Groq)':<35} {bl_accuracy:>8.3f}")
print(f"{'Fine-tuned DistilBERT':<35} {ft_accuracy:>8.3f}")
print("-" * 45)
delta = ft_accuracy - bl_accuracy
direction = "improvement" if delta >= 0 else "regression"
print(f"\nFine-tuning {direction}: {abs(delta):.3f}")
print()
print("Use these numbers in your README evaluation report.")
```

```
==================================================
RESULTS COMPARISON
==================================================
Model                                Accuracy
---------------------------------------------
Zero-shot baseline (Groq)               0.879
Fine-tuned DistilBERT                   0.364
---------------------------------------------

Fine-tuning regression: 0.515

Use these numbers in your README evaluation report.
```

```
# Save results JSON — commit to your GitHub repo and reference in README
results = {
    "baseline_accuracy": round(bl_accuracy, 4),
    "finetuned_accuracy": round(ft_accuracy, 4),
    "improvement": round(ft_accuracy - bl_accuracy, 4),
    "test_set_size": len(test_df),
    "label_map": LABEL_MAP,
    "model": MODEL_NAME,
}
with open("evaluation_results.json", "w") as f:
    json.dump(results, f, indent=2)

print("✅ Files ready to download:")
print("   evaluation_results.json  — metrics for your README")
print("   confusion_matrix.png     — include in your README")
print()
print("Download via: Files panel (📁) on the left → right-click → Download")
```

```
✅ Files ready to download:
   evaluation_results.json  — metrics for your README
   confusion_matrix.png     — include in your README

Download via: Files panel (📁) on the left → right-click → Download
```